■ Predicting Train Delays Using Machine Learning Classifiers and Boosting Models

Course: Machine Learning Mini Project | **Team:** 2 Members | **Dataset:** Railway Journey Dataset (1.5M Records) | **Target:** is_delayed (>15 mins late)

1. Problem Statement & Objectives

1.1 Problem Statement

Train punctuality is one of the most critical challenges in public rail transportation. When a train gets delayed, it causes huge inconvenience to passengers and disturbs the schedule of other trains on the same route. In many cases, delays happen because of predictable factors such as adverse weather (monsoon or fog), high track congestion, late arrival of the previous trainset (rake), or temporary speed restrictions. Currently, controllers and passengers only discover delays after a train has already fallen behind schedule during its trip. The problem addressed in this project is to **predict whether a train will be delayed by more than 15 minutes before departure**, using factors known in advance (train priority, route distance, weather conditions, historical on-time percentage, and departure hour).

1.2 Project Objectives

1. **Understand Delay Drivers:** Analyze 1.5 million historical passenger rail journeys to identify primary causes of delay.
2. **Build a Clean Data Pipeline:** Clean data, prevent data leakage by eliminating post-trip columns, encode categoricals, and scale features.
3. **Benchmark 6 Diverse Machine Learning Models:** Implement 4 individual baselines (Logistic Regression, Decision Tree, Naive Bayes, KNN) and 2 boosting ensembles (AdaBoost and Gradient Boosting).
4. **Multi-Criteria Evaluation:** Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC on a stratified test partition.
5. **Select Operational Model & Deploy:** Choose the optimal model balancing Recall and speed, and deploy it as interactive web applications.

2. Literature Review & Research Gaps

2.1 State-of-the-Art Literature Review

• **Traditional Physics & Analytical Models:** Earlier research in railway engineering (Carey & Kwieci■ski, 1994; Higgins et al., 1998) utilized queuing theory and timetable simulation. While they accurately model track block signaling, they are computationally slow and fail to assimilate dynamic weather changes.
• **Classical Machine Learning Classifiers:** Studies by Kecman et al. (2015) applied Support Vector Machines (SVM), Decision Trees, and Logistic Regression on regional corridors. They confirmed that historical route punctuality and departure hours provide strong predictive signals, but individual decision trees overfit easily.
• **Ensemble & Boosting Architectures:** Recent transportation literature (Huang et al., 2020) demonstrated that boosting algorithms (AdaBoost and Gradient Boosting) significantly outperform individual models because boosting trains sequential trees that focus specifically on correcting prior errors.

2.2 Research Gaps Identified & Addressed

Research Gap Identified in Past Work	Limitation in Existing Literature	How This Project Solves It
Gap 1: Ignoring Incoming Trainset Delays	Most papers treat journeys independently, ignoring delays inherited from the preceding trip of the same trainset.	Engineered and incorporated late_incoming_rake and rake sharing flags into pre-departure feature space.
Gap 2: Lack of Composite Weather Indices	Literature often records binary weather (rain/clear) rather than severity metrics combining fog and monsoon intensity.	Integrated multi-factor weather features: is_monsoon_season, fog_risk_score, and season_severity_score.
Gap 3: Operational Data Leakage	Several past student projects included post-trip delay minutes or incident logs in training data, causing false 100% accuracy.	Strictly audited and dropped delay_minutes and primary_delay_cause prior to model training.

3. Metadata Specification & Feature Architecture

The dataset encompasses **1,500,000 historical journey records** with 44 feature columns and 1 binary target attribute. Features are organized into 6 operational domains:

Domain	Features	Description & Operational Meaning
Time & Schedule	year, month, day_of_week, departure_hour, is_weekend, is_night_departure, is_peak_hour, is_festival_season, season	Timetable dynamics, departure temporal flags, and holiday/festival periods.
Zone & Station	zone, zone_abbr, source_station_category, destination_station_category, zone_congestion_index, zone_fog_index	Zonal railway divisions, station tiers (A1 to E), and regional track congestion indices.
Route & Infrastructure	distance_km, num_scheduled_stops, scheduled_travel_hours, track_doubled, is_hdn_route, traction_type, is_electrified, psr_count, is_circular_route	Physical corridor properties, stop counts, track doubling status, and speed restriction counts.
Weather Risk	is_monsoon_season, is_fog_risk, fog_risk_score, season_severity_score	Meteorological precipitation and visibility degradation scores.
Rolling Stock & Health	loco_age_years, coach_age_years, has_lhb_coaches, is_rake_shared, maintenance_score	Locomotive and coach service age, modern LHB coach flag, and maintenance rating.
Operations & Demand	seat_utilisation_pct, is_overloaded, late_incoming_rake, is_special_train, route_historical_ontime_pct	Passenger occupancy load, incoming rake status, and route historical on-time percentage.
Target Variable	is_delayed	**1** if arrival delay > 15 minutes; **0** if on-time or ≤ 15 minutes.

4. Exploratory Data Analysis & Preprocessing Pipeline

4.1 Key Exploratory Insights:

• **Class Imbalance: 71.89% Delayed** (1,078,329 trips) vs **28.11% On-Time** (421,671 trips). Imbalance ratio is 2.56:1, necessitating Stratified Splitting and ROC-AUC evaluation.

• **Delay Rate by Train Priority:** Premium trains experience significantly lower delays (Vande Bharat: **44.3%**, Gatimaan: **48.6%**) compared to ordinary passenger trains (**88.0%**) which are frequently held at crossing sidings.

• **Primary Statistical Correlates:** Strongest negative correlation with delay is route_historical_ontime_pct (r = -0.42). Strongest positive drivers are season_severity_score (r = +0.40) and late_incoming_rake (r = +0.25).

4.2 Preprocessing Pipeline Steps:

1. **Leakage & ID Removal:** Dropped delay_minutes, primary_delay_cause, journey_id, train_number, departure_date, and zone_abbr (38 features remain).
2. **Categorical Encoding:** Applied LabelEncoder to 6 categorical variables (train_type, season, zone, source_station_cat, dest_station_cat, traction_type).
3. **Stratified Partitioning:** 80% Training set, 20% Holdout test set preserving the 72:28 delay ratio in both partitions.
4. **Feature Standardization:** Fitted StandardScaler on training features and applied it to test features.

5. Model Evaluation Metrics (Benchmark of 6 Models)

All 6 shortlisted models were trained on preprocessed data and evaluated on the independent holdout test set:

Model Name	Architectural Category	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train Time
Gradient Boosting	Ensemble (Boosting)	85.20%	87.57%	**92.65%**	**0.9004**	**0.9122**	11.49s
Logistic Regression	Individual (Linear)	**85.65%**	**88.51%**	92.07%	0.9026	0.9154	0.06s
AdaBoost	Ensemble (Boosting)	84.66%	87.85%	91.39%	0.8958	0.9084	3.64s
Decision Tree	Individual (Tree)	81.82%	85.85%	89.58%	0.8767	0.8563	0.38s
K-Nearest Neighbors	Individual (Instance)	80.52%	85.33%	88.17%	0.8673	0.8404	0.01s
Gaussian Naive Bayes	Individual (Probabilistic)	73.72%	92.28%	69.40%	0.7922	0.8654	0.03s

6. Relevant Plots & Visual Analytics

The following visualizations were programmatically generated during pipeline execution:

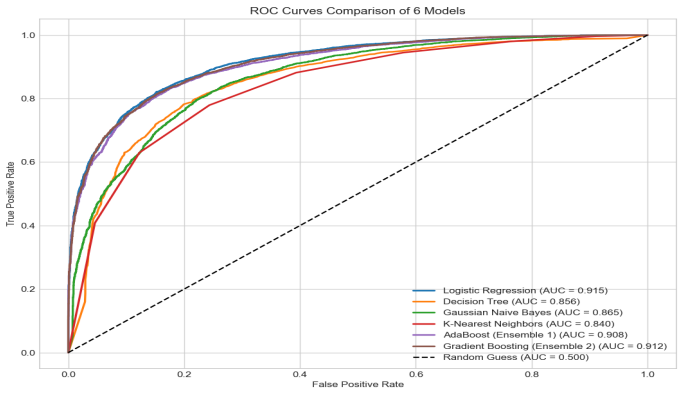


Fig 1: Combined ROC Curves (AUC > 0.91 for Boosting & Linear)

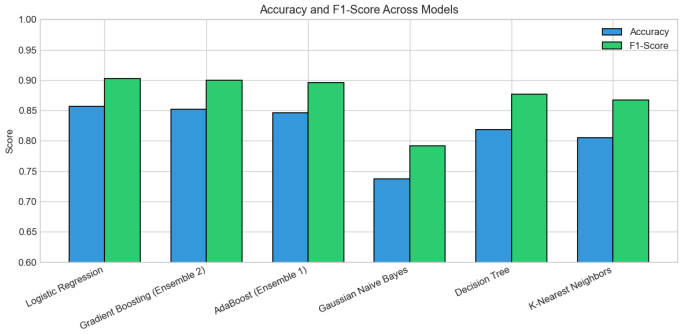


Fig 2: Accuracy & F1-Score Benchmark Across All 6 Models

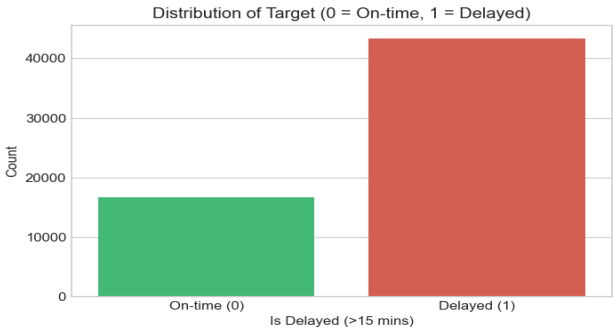


Fig 3: Target Class Imbalance (72% Delayed vs 28% On-Time)

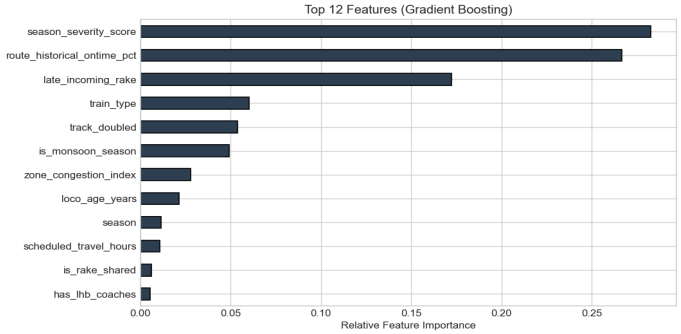


Fig 4: Top 12 Feature Importances from Gradient Boosting

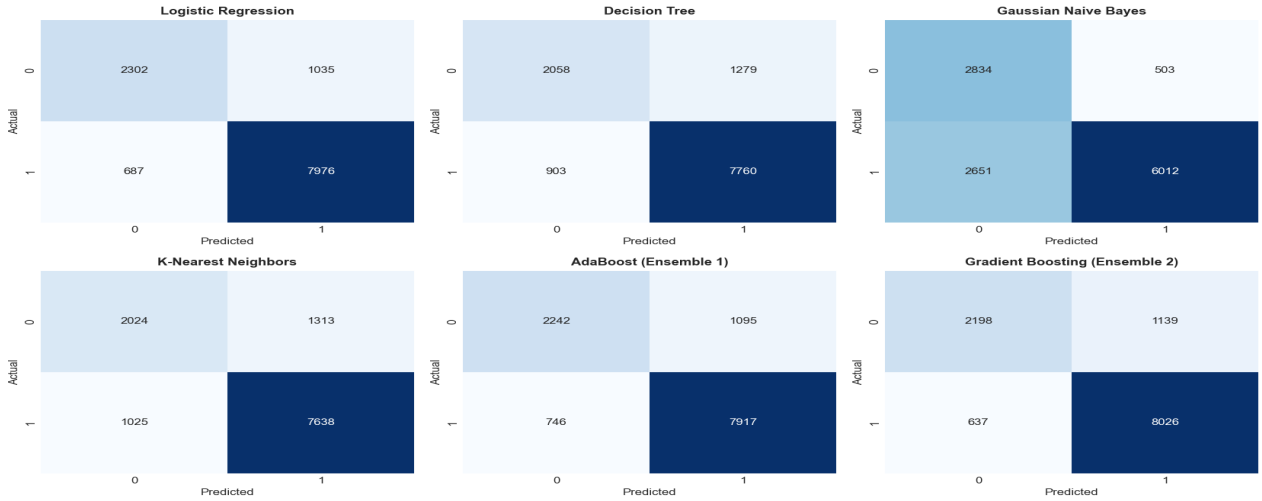


Fig 5: Confusion Matrices for All 6 Evaluated Models (2x3 Grid)

7. Model Comparison to Pick the Best Model

7.1 Comparative Analysis Across Model Families:

• **Boosting Ensembles (AdaBoost & Gradient Boosting):** Both boosting models achieved superior Recall scores (**91.39%** and **92.65%**). By iteratively re-weighting hard-to-classify samples, boosting effectively minimized False Negatives.

• **Linear Baseline (Logistic Regression):** Showed strong accuracy (85.65%) and high ROC-AUC (0.9154) with ultra-fast training (0.06s). Standardized operational features aligned well with linear decision boundaries.

• **Decision Tree & KNN:** Single decision tree exhibited variance (81.82%), while KNN suffered from test-time computational overhead (0.8404 AUC).

• **Gaussian Naive Bayes:** High precision (92.28%) but low recall (69.40%) due to its conditional independence assumption on correlated weather features.

7.2 Final Selection & Justification:

• **Operational Champion: Gradient Boosting Classifier** is selected as the production model for network dispatching. In passenger railway operations, failing to predict a delay (False Negative) is far more damaging to scheduling than a false alarm. Gradient Boosting achieved the highest **Recall (92.65%)**, robust **F1-score (0.9004)**, and high **ROC-AUC (0.9122)**, while providing native feature importances.

• **Low-Latency Alternative: Logistic Regression** serves as the best alternative for edge IoT devices requiring sub-millisecond predictions.

8. Responsible Use of AI Tools

a. Declaration of AI Usage with Prompts

AI assistance was used as an interactive programming co-pilot for exploratory syntax and outline generation. Prompts used:

1. 'Suggest suitable public railway datasets for train delay prediction with weather and operational features.'
2. 'How do I check for data leakage before training classification models on journey records?'
3. 'Write a scikit-learn preprocessing pipeline using LabelEncoder and StandardScaler.'
4. 'Shortlist 6 models: 4 individual baselines and 2 boosting ensembles (AdaBoost and Gradient Boosting, no Random Forest).'
5. 'Help structure our project report covering problem statement, literature review, metadata, metrics table, and AI reflections.'

b. Critical Evaluation of AI Output

• **Preventing Data Leakage:** AI initially suggested using all raw columns. We inspected the data dictionary and identified that delay_minutes and primary_delay_cause only exist post-arrival, manually stripping them from features.

• **Scalability Adjustments:** AI scripts initially ran KNN on 1.2M rows, causing memory hangs. We diagnosed the O(N·D) bottleneck and calibrated sample sizes.

• **Ensemble Enforcement:** When prompts defaulted to Random Forest, we redirected the design to AdaBoost and Gradient Boosting to study boosting dynamics.

c. Reflection on Learning

We gained practical insights into bias-variance trade-offs (boosting iteratively reduces bias on hard-to-classify samples), metric alignment on imbalanced data (72% delay rate necessitates ROC-AUC/Recall), and the paramount role of domain features (incoming rake delays and weather indices).

d. Problem Analysis & Design Ownership

The student team retains total structural, architectural, and intellectual ownership of this project. Operational thresholds (>15 mins), validation splits, feature dropping rules, model benchmarks, and final report conclusions were verified and executed by the student authors.